

Education

- Sep 2024 – Technion - Israel Institute of Technology, *PhD, Faculty of Data and Decision Sciences*.
Advisor: Noa Zychlinski
- Sep 2021 – Shanghai University, *Master, Management Science and Engineering*.
Apr 2024
- Sep 2017 – Harbin University of Science and Technology, *Bachelor, Statistics*.
Jun 2021

Research Interests

I am interested in operations management and operations research in healthcare because they use quantitative mathematical tools, such as queueing theory and stochastic modeling, to improve efficiency, access, and quality of care under limited resources and uncertainty.

Published Papers

'*' means the corresponding author.

- 7 **Cai J***, Likvornik N.F, Zychlinski N. From waitlists to well-being: An analytical approach to AI-supported mental healthcare, *IISE Transactions*, *forthcoming*.
- 6 **Cai J***, Zychlinski N. When AI is not enough: Reducing diagnostic errors with radiologist oversight, *Service Science*, *forthcoming*.
- 5 **Cai J***, Zychlinski N. Expanding naloxone accessibility: A lifesaver or a risky setback?, *Production and Operations Management*, 2026, 35(6):2376-2394..
 - Winner of the 2025 Doctoral Student Forum of Antai College of Economics and Management, Shanghai Jiao Tong University, July 10, 2025.
- 4 Zhou J, **Cai J***, Pantelous A, Li Z, Li M. A decision-making framework for supporting an equitable global vaccine distribution under humanitarian perspectives, *European Journal of Operational Research*, 2025, 327(2):655-672.
 - Winner of the 2024 Doctoral Student Forum of Antai College of Economics and Management, Shanghai Jiao Tong University, July 11, 2024.
- 3 **Cai J**, Xu R, Zhou J*, Gu H. Dynamic-spread assessment and kink detection in the early mpox outbreak, *Applied Mathematical Modelling*, 2025, 143:115996.
- 2 **Cai J**, Zhou J*. How many asymptomatic cases were unconfirmed in the US COVID-19 pandemic? The evidence from a serological survey, *Chaos, Solitons and Fractals*, 2022, 164:112630.
- 1 Li H, **Cai J***. Arithmetic operations and expected values of regular interval type-2 fuzzy variables, *Symmetry-Basel*, 2021, 13(1):2196.

Selected Honors

- 2022 National scholarship for graduates.